WORKING NOTE · JUNE 2026

Share-faithful reconstruction of bilateral trade from raw UN Comtrade — and a pre-registered nowcast for critical materials

Varcolacus · independent research · public data only · comtrade-reconcile · interactive atlas

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Abstract. I reconstruct bilateral trade in 32 critical raw materials from raw UN Comtrade using a Gaulier–Zignago-style mirror reconciliation (reliability-weighted inverse-variance averaging of exporter and importer reports), and validate it against the official CEPII BACI dataset on the statistics that matter for supply-concentration analysis: top exporters, trade shares, and Herfindahl concentration. On 2024 the reconstruction recovers the top exporter in **25 of 30** materials with a share mean-absolute-error of **3.5pp** and an HHI correlation of **0.92** (importer side: 22/30, 4.2pp, 0.97). I then (i) document an **origin gap** — in **19 of 32** materials the top *exporter* is not the top *miner*, so import-origin statistics systematically overstate the geographic diversification of supply — and (ii) build a provisional 2025 nowcast and a directional 2026 scenario whose shared assumption (share persistence) I test out-of-sample (**85%** year-over-year top-exporter persistence; a leader's annual share move has P50 3.5pp / P90 8.7pp) and whose accuracy I **pre-register** against the next BACI release. All inputs are public; the raw Comtrade is committed, so all code and fixtures reproduce the reconciliation and validation end-to-end with no API key.

1. The question

Under customs rules, refining and substantial transformation confer "origin": when a country imports an ore, processes it and re-exports, the traded good takes the processor's nationality and the customs record stops there. Trade statistics therefore measure *where a material was last shipped from*, not *where it came out of the ground*. For critical-material supply-risk analysis — where the policy question is genuine geographic concentration of the upstream — this is a systematic blind spot. This note builds the data to measure it and quantifies how large the gap is.

2. Data and vintages

Table 1 — sources and exact vintages. All public.

LAYER	SOURCE	VINTAGE / COVERAGE
Reconciled bilateral trade (reference)	CEPII BACI	HS17 release **V202601** (Jan 2026), years 2018–2024
Raw bilateral trade (reconstruction input)	UN Comtrade API	annual + monthly, reporter×partner, pulled 2025–26
Mine production shares	USGS Mineral Commodity Summaries	latest published (approx.)
Refining / processing shares	IEA Critical Minerals Outlook	latest published (approx.)
Reserves	USGS Mineral Commodity Summaries	economically recoverable (approx.)
Gravity covariates (CIF/FOB)	CEPII `dist_cepii`	distance, contiguity
Commodity prices (2026 tilt)	World Bank Pink Sheet	monthly, Q1-2026 vs Q1-2025

The 32 materials are mapped to HS6 product codes; three of them — gallium, germanium and hafnium — share a single HS6 code (811292) and so carry *identical* trade columns that cannot be separated. This is flagged wherever those materials appear.

3. Reconciliation method

For each product and country pair *(i, j)* two reports exist: the exporter's free-on-board value x_{ij} and the importer's cost-insurance-freight value m_{ji}. Following Gaulier & Zignago (2010), I (a) strip an estimated CIF/FOB markup from import values, (b) weight each reporter by an estimate of its reliability, and (c) average the two reports as an inverse-variance weighted mean in logs.

CIF/FOB markup

The markup is recovered from the within-product mirror ratio. On the 32-material slice a gravity regression of $\log(m/x)$ on distance and contiguity is not identified ($R^2 \approx 0.01$) — BACI estimates this on the full ~5,000-product universe — so I fall back to a robust per-product median markup, the same quantity in reduced form.

Reliability weights (variance components)

Let $d_{ij} = \log x_{ij} - \log m_{ji}$ be the mirror discrepancy. Modelling it as the sum of a reporter-i and reporter-j error, an OLS regression of d^2 on reporter dummies yields each reporter's error variance v_i (a variance-components estimator). The reconciled value averages the two reports with weights inversely proportional to those variances:

$$\log t_{ij} = (w_i \cdot \log x_{ij} + w_j \cdot \log \tilde{m}_{ji}) / (w_i + w_j), \quad w_i = 1 / v_i$$

where $\tilde{m}$ is the import value after removing the CIF/FOB markup. The result is a single reconciled bilateral matrix per product per year — the same object BACI produces, built here from the raw reports.

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4. Validation against BACI

The reconstruction is validated on the statistics the atlas actually uses — shares, ranks, concentration — not on absolute levels.

Table 2 — reconstruction vs official BACI, both sides of the market.

YEAR · SIDE	TOP-1 EXPORTER/IMPORTER CORRECT	SHARE MAE	HHI CORRELATION
2024 · exporter	25 / 30	3.5 pp	0.92
2024 · importer	22 / 30	4.2 pp	0.97
2022 · exporter	22 / 30	3.9 pp	0.89

On levels. Reconstructed totals run ~1.5–1.8× BACI. This is *not* a reconciliation artefact: current raw Comtrade already sits ~1.8–1.9× above BACI's published snapshot before any reconciliation (Comtrade absorbs continuous revisions; BACI applies additional outlier/quality filtering this note does not replicate), and the reconstruction sits *between* the two raw mirror reports. A settled year (2022) still shows ~1.5×, so the offset is partly persistent filtering, not only vintage. The claim is therefore explicitly **share-faithful**, not level-faithful — which is the relevant property for concentration analysis.

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5. Nowcast and pre-registration

BACI lags ~18 months. For the years it has not released I extend the series:

- **2025 (provisional).** The full reconciliation applied to the partial 2025 Comtrade that has been filed (~half of reporters by mid-2026), with levels calibrated per material to BACI 2024.

- **2026 (directional).** Only ~Q1 monthly Comtrade exists, so 2025's reconciled structure is carried forward and only *levels* are tilted, per material, by reporter-matched Q1 export momentum blended with the World Bank Pink Sheet price change. Shares are held at 2025: this is a directional scenario, not a bilateral measurement.

Both rest on one assumption — that last year's trade structure predicts this year's. I test it directly on the measured 2018–2024 series, scoring year $T-1$'s shares as a nowcast for year T:

Table 3 — out-of-sample persistence (192 material-years, 2018–2024).

METRIC	VALUE
Top-exporter unchanged year-over-year	85%
Mean share MAE across exporters	0.37 pp
Leader's annual share move — median (P50)	3.5 pp
Leader's annual share move — P90	8.7 pp

The P50/P90 are the nowcast's empirical uncertainty band, shown on the interactive year slider. Most-predictable materials: magnets, coking coal, manganese, niobium; least: arsenic, beryllium, hafnium, gallium, germanium, fluorspar (thin or non-reporting-producer markets).

Pre-registration. Before BACI 2025 exists, I have committed (`PREREGISTRATION.md`, in git history at the relevant commit) to how the frozen `flows_2025.json` will be scored when it releases — `validate.py 2025`, same metrics — with numeric thresholds (≥ 22/30 top-1, ≤ 5pp share MAE, ≥ 0.88 HHI correlation; scored on the 30 validatable materials) and a commitment to publish the result pass or fail. The bet is written down first; the test is falsifiable on a fixed date by one command.

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6. Finding: the origin gap

With reconciled trade and the mine-production layer side by side, define, per material in a given year:

$$\textit{origin gap} = (\textit{top exporter's share of world trade}) - (\textit{that same country's share of world mine output})$$

A large positive gap means a country sells far more than it mines — it processes or trans-ships someone else's ore. The headline result, on 2024: in **19 of 32** materials the top exporter is not the top miner; in **4 of 32** a country mining under 5% of world supply exports more than a quarter of it.

Table 4 — largest origin gaps, 2024 (where exporter ≠ miner).

MATERIAL	TOP EXPORTER	EXPORTS	IT MINES	GAP	LEAD MINER
Beryllium, unwrought	Kazakhstan	89%	0%	+89	United States (65%)
Strontium carbonate	Germany	60%	0%	+60	Spain (30%)
Lithium carbonate	Chile	75%	22%	+53	Australia (52%)
Bauxite / aluminium ore	Guinea	72%	25%	+47	Australia (27%)
Cobalt oxides & hydroxides	Finland	29%	0%	+29	DR Congo (76%)
Tantalum, unwrought	United States	22%	0%	+22	DR Congo (40%)
Nickel, unwrought	Norway	18%	0%	+18	Indonesia (60%)

The twist. The intuitive story is "China hides behind refineries." The data only half-supports it: for many materials China is *both* the lead miner and the lead exporter — its position is largely genuine, not an artefact. The materials where exporter and miner diverge are instead fronted by industrial refiners and entrepôts — Finland for Congolese cobalt, Japan for largely Chinese-mined titanium, Germany for Spanish strontium, Norway for Indonesian nickel. The corrective therefore cuts both ways: it deflates apparent dependence on refiner countries, and it reveals that genuinely concentrated upstreams (Congo cobalt, Indonesian nickel) are *more* concentrated than the diversified-looking trade ledger suggests.

7. Limitations and non-claims

- The origin gap is a **measurement gap** between two public datasets (reconciled trade shares vs USGS mine shares), *not* a fully traced physical supply chain. Some gap is legitimate — a refiner genuinely exports a *different product* than the mined ore (cobalt chemicals vs cobalt ore), which sits under a different code.
- Customs data cannot separate a true refiner from a re-export hub; trade-derived country roles are *trade exposure*, not proof of physical processing. Entrepôts are flagged.
- gallium / germanium / hafnium share one HS6 code and cannot be separated in trade.
- Figures are share-faithful, not level-faithful; absolute values are not claimed. Mine and refine layers are approximate USGS/IEA reference shares, undated. 2025 is provisional; 2026 is directional.

8. Reproducibility

Two public repositories, **no API key required to reproduce the pipeline end-to-end**: the interactive atlas (Varcolacus/critical-materials-atlas) and the reconciliation engine (Varcolacus/comtrade-reconcile, CI green). The raw 2024 Comtrade is committed (gzipped) under `fixtures/raw/`, so the reconciliation regenerates from raw rather than from a pre-computed file. On a fresh clone of the engine: `pip install -r requirements.txt`, then `ATLAS_ROOT=fixtures python reconcile.py 2024` (regenerates the reconciliation), `… validate.py 2024` (Table 2), `python backtest.py` (Table 3), `python findings.py` (Table 4). CI runs the `reconcile → validate` chain on every push. The Comtrade key is needed only for the initial network pull and is read from the environment, never committed.

References

- Gaulier, G. & Zignago, S. (2010). *BACI: International Trade Database at the Product-Level*. CEPII Working Paper 2010-23.
- UN Comtrade Database. United Nations Statistics Division.
- U.S. Geological Survey. *Mineral Commodity Summaries*.
- International Energy Agency. *Global Critical Minerals Outlook*.
- World Bank. *Commodity Markets ("Pink Sheet")*.